

AYA DALACHE

AI / Machine Learning Engineer • Backend & Full-Stack Software Engineer(Web)

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PROFESSIONAL SUMMARY

Software engineer with a Master's-level research focus on AI safety, NLP, and applied machine learning, combined with hands-on experience building production backend systems. Fine-tuned and evaluated transformer models for Arabic dialect-aware content moderation, and engineered end-to-end information retrieval and full-stack platforms using Django, Laravel, and REST APIs. Comfortable operating across the stack: model training and evaluation, database design and performance benchmarking (PostgreSQL, Oracle SQL), and modern frontend engineering with vanilla JavaScript and Three.js. Builds systems that are measured, benchmarked, and interview-defensible rather than feature lists.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), PHP, Java, SQL, R, Bash, C++

Web Development: Django, Django REST Framework, Laravel, HTML5, CSS3, Tailwind CSS, Bootstrap 5, Three.js, REST APIs

AI & Machine Learning: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LangChain, LangGraph

Natural Language Processing: Arabic NLP, Information Retrieval, OCR, TF-IDF, Transformer Fine-Tuning

Databases: PostgreSQL, MongoDB, MySQL, Oracle SQL, PL/SQL, SQLite, Database Design, Data Mining, Data Warehousing

DevOps & Tools: Docker, Git, GitHub, Linux, Composer, npm, Vite

Software Engineering: OOP, MVC, Design Patterns, Data Structures & Algorithms, Clean Code, Unit Testing

Data Analysis & BI: Power BI

PROFESSIONAL EXPERIENCE

Web & Software Development Intern — Intellectsoft

Sétif, Algeria — 2-Month Internship

- Collaborated within a professional software development team in a production-oriented, industry environment.
- Participated across the full software development lifecycle: requirements analysis, application development, testing, and deployment.
- Strengthened full-stack development skills through collaborative version-control workflows using Git.
- Applied software engineering best practices and Agile development principles in a live delivery setting.

SELECTED PROJECTS

AI, Machine Learning & NLP:

AMAN-LLM — Ethical Constraint Enforcement for Tool-Using Arabic LLM Agents (Master's Thesis)

(github.com/ayadalache/aman-llm-agent)

Python • PyTorch • Transformers • LangChain • LangGraph

- Designed a multi-layer AI safety framework for Arabic-speaking LLM agents, combining harmful-content detection, policy enforcement, and constitutional self-critique.
- Fine-tuned MARBERT and AraBERT models for dialect-aware content moderation, achieving 89.2% accuracy and 91.6% F1-score.
- Built a 500-case adversarial benchmark spanning Modern Standard Arabic, Algerian Darja, and Arabizi to stress-test prompt-injection and jailbreak resistance.

Arabic Fiqh Information Retrieval System *(github.com/ayadalache/arabic-fiqh-ir-system)*

Python • Streamlit • Transformers • OCR

- Engineered an end-to-end Arabic search engine integrating OCR, Arabic NLP preprocessing, positional indexing, TF-IDF ranking, and transformer-based text classification.

- Implemented Boolean, phrase, proximity, and ranked retrieval modes, exposed through an interactive Streamlit interface.

Face Recognition System (github.com/ayadalache/PCA-Face-Recognition)

Python • OpenCV • Scikit-learn

- Developed an end-to-end face recognition pipeline using PCA-based feature extraction and supervised machine learning classifiers.

Machine Learning Regression Projects

Python • PyTorch

- Built a collection of end-to-end regression models covering housing-price prediction and medical-cost estimation, including data preprocessing, exploratory analysis, model development, and evaluation.

Full-Stack Web Applications:

BT Real Estate Platform (github.com/ayadalache/BT-Real-Estate-Full-Project)

Django • Django REST Framework • JWT • REST APIs

- Engineered a full-stack real estate platform with secure JWT authentication, RESTful APIs, and role-based access control (RBAC) for customer and staff dashboards.
- Implemented advanced property search and inquiry-management workflows across separate customer and staff-facing interfaces.

Sift — Task Management Application (github.com/ayadalache/sift-real)

Laravel • Tailwind CSS

- Developed a Laravel-based task management application with secure authentication, ownership-based authorization, and category management.
- Built a dashboard analytics view and a custom Tailwind CSS design system.

Virtual Chemistry Lab 3D (github.com/ayadalache/Virtual-Chemistry-Lab-3D)

Three.js • WebGL • PHP • MySQL

- Engineered a full-stack educational web application simulating chemistry experiments in an interactive 3D laboratory built with Three.js and WebGL.
- Implemented role-based access, experiment recording, submission workflows, and instructor grading on a PHP/MySQL backend.

TasteBite — Desktop Recipe Manager (github.com/ayadalache/tastebite)

Python • Kivy

- Built a desktop recipe management application using a custom binary search tree for efficient recipe indexing.
- Implemented advanced search, personalized recommendations, and persistent local storage.

Systems, Performance & Database Engineering

Ledger — Framework-Free JavaScript Task Manager (github.com/ayadalache/Ledger)

JavaScript • IndexedDB • Web Workers

- Developed a task management application in vanilla JavaScript, implementing a command-pattern undo/redo engine and custom reactive state management without a framework.
- Optimized runtime performance using virtualized rendering, Web Workers, and IndexedDB persistence.

TPC-H PostgreSQL Benchmark & Partitioning (github.com/ayadalache/tpch-postgresql-partitioning-project)

PostgreSQL • Bash • Linux

- Implemented and evaluated multiple PostgreSQL table-partitioning strategies against the TPC-H benchmark dataset.
- Benchmarked analytical SQL workload performance across partitioning schemes on Linux.

Additional Projects

Quran & Bible Search Engine • Fraud Detection Model • Malicious URL Detection • Spam Classification • Personal Portfolio Website

EDUCATION

CERTIFICATIONS

- Web Development Diploma (International Gold Diploma – Level 17), Meshkat Educational Foundation
- Front-End Development, Tutori Land Online Academy
- Django Framework Development
- Laravel Framework Development
- Data Analysis, Machine Learning & Deep Learning
- Power BI Data Analysis & Visualization
- AI-Assisted Graphic Design
- Social Media Marketing Fundamentals

WORKSHOPS & PROFESSIONAL DEVELOPMENT

- Participated in workshops and technical training covering web development, digital marketing, artificial intelligence, Power BI, and WordPress.
- Attended a conference on digital investment and online revenue generation organized by Al-Quds Business.
- Completed hands-on training in WordPress and Content Management Systems (CMS).

LANGUAGES

Arabic — Native | English — B1 (Professional Working Proficiency) | French — B1 (Professional Working Proficiency)